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Studierichting: Informatica
Oefeningenreeks 1D nr. 24.2

Opgave: Los de volgende vergelijking op in $\mathbb{C}$

$$8z^6 + 7iz^3 + 1 = 0$$

Stel $w = z^3$.

$$8w^2 + 7iw + 1 = 0$$

$$D = (7i)^2 - 4 \cdot 8 \cdot 1 = -81$$

Resolvente:

$$\begin{cases} x^2 - y^2 = -81 \\ 2xy = 0 \Rightarrow x^2(-y^2) = 0 \end{cases}$$

$$\lambda^2 + 81\lambda = 0 \Rightarrow \lambda_1 = 0, \lambda_2 = -81$$

$$x^2 = 0 \Rightarrow x = 0 \text{ en } y^2 = 81 \Rightarrow y = 9$$

$$w_1 = \frac{-7i + 9i}{16} = \frac{1}{8}i \text{ en } w_2 = \frac{-7i - 9i}{16} = -i$$

$$\Rightarrow z^3 = \frac{1}{8}i \text{ en } z^3 = -i$$

oplossingen:

voor w_1

$$\left| \frac{1}{8}i \right| = \frac{1}{8}$$

$$\theta_1 = \arg(w_1) = \frac{\pi}{2}$$

$$z_k = \sqrt[3]{\frac{1}{8}} \left(\cos \frac{\frac{\pi}{2} + 2k\pi}{3} + i \sin \frac{\frac{\pi}{2} + 2k\pi}{3} \right)$$

$$k = 0 \Rightarrow z_0 = \frac{1}{2} \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right) = \frac{\sqrt{3}}{4} + \frac{1}{4}i$$

$$k = 1 \Rightarrow z_1 = \frac{1}{2} \left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6} \right) = -\frac{\sqrt{3}}{4} + \frac{1}{4}i$$

$$k = 2 \Rightarrow z_2 = \frac{1}{2} \left(\cos \frac{9\pi}{6} + i \sin \frac{9\pi}{6} \right) = \frac{1}{2}(0 - i) = -\frac{1}{2}i$$

voor w_2

$$|-i| = 1$$

$$\theta_2 = \arg(w_2) = \frac{3\pi}{2}$$

$$z_k = \sqrt[3]{1} \left(\cos \frac{\frac{3\pi}{2} + 2k\pi}{3} + i \sin \frac{\frac{3\pi}{2} + 2k\pi}{3} \right)$$

$$k = 0 \Rightarrow z_3 = \cos \frac{\pi}{2} + i \sin \frac{\pi}{2} = i$$

$$k = 1 \Rightarrow z_4 = \cos \frac{7\pi}{6} + i \sin \frac{7\pi}{6} = -\frac{\sqrt{3}}{2} - \frac{1}{2}i$$

$$k = 2 \Rightarrow z_5 = \cos \frac{11\pi}{6} + i \sin \frac{11\pi}{6} = \frac{\sqrt{3}}{2} - \frac{1}{2}i$$

References